

Hubble Classification System:

- Stars in MB1:

→ 4 types of galaxies

- (i) Spiral
- (ii) Elliptical
- (iii) Lenticular
- (iv) irregular

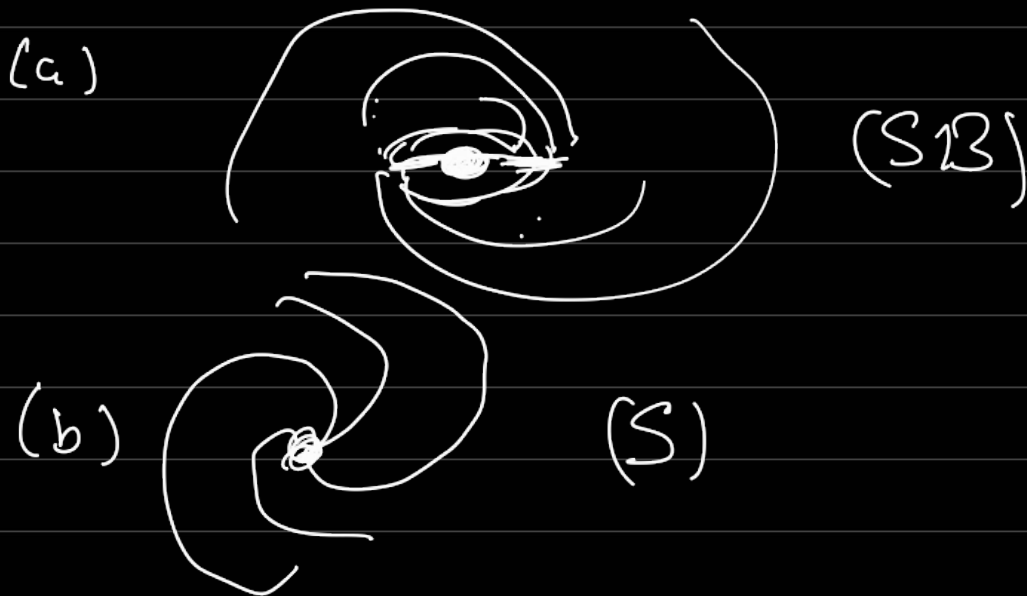
(i) • Extended disks

- Central Bulge

- Have equal proportions of

(a) Spiral Barred (SB) $\sim 50\%$

(b) Spiral (unbarred) (S) $\sim 50\%$



- (Bulge/disk) $\sim$ correlate $\sim$ $\left(\frac{\text{arm tightness}}{\text{lumpiness}} \right)$

Sa, ab, b, b, c, cd, d

(Sa): have largest (B/D) ratios and most tight/
barred, smooth spiral arms.

• MW: $\sim$ Sc

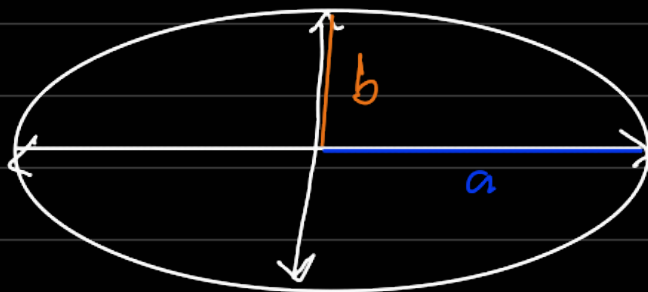
M31: Sb

Elliptical: (spheroidal)



• Classify by ellipticity E_n :

$$n \equiv 10 \left(1 - \frac{b}{a} \right)$$



* $E0$: $n=0 \Rightarrow 1 - \frac{b}{a} = 0 \Rightarrow a=b$
Circle

• More concretely we can look at shapes & velocities (Binney (1985)):

Triaxial Ellipsoids: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$



• Supported against gravity through random motion of stars.

• Small ell. rotate

• Generally clustered together

• Most luminous galaxies known are CD
(Cluster dominant)

$$L_{CD} \sim 100 L_{MW}$$

• Hubble Profile: $\sum_{1 \leq i} (R) = \frac{\sum_{1 \leq i} 1}{(1 + \frac{R}{a})^2}$

• de Vaucouleurs: $\sim -\kappa R^{1/4}$

profile

$$\Sigma_{de}(R) = \sum_0 e^{-kR}, k \in \mathbb{R}$$

- Smooth, no gas or dust
- Red: only old stars are
- Dynamically old

• Many obs. show these forming from merging spirals.

Spiral: $a \rightarrow c$:

- openness of spiral arms increases
- resolution of arms into clumps incre.
- Size of bulge decreases

• $I(R) \sim \exp(-R/R_0)$

Lenticulars: Between E & S

* Called S0's

- Large bulge ⊕ disk
- No spiral structure
- Barred or unbarred: (S0/SB0)
- Found in dense regions

Irregular: Not symmetric "eccentric/random looking"

Irrs

- Low Luminosity
 - Blue $\rightarrow$ star forming
- Large Magellanic Cloud is an example

Dwarf Galaxies: fainter than $(L \approx 5 \times 10^9 L_\odot)$
 $M_V \leq -17$

(i) Depend on band used for observing.

B: Young Stars

→ often clustered in patchy distributions

→ look irregular

R: older Stars

→ smoother dist.

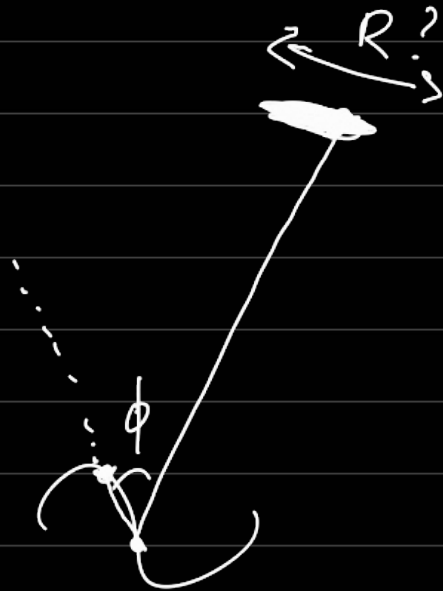
(ii) Resolution

(iii) Depth of exposure

(iv) Apparent $\neq$ intrinsic

Surface Brightness profile:

Surface Brightness: $I(R, \phi) \equiv$ amount of light
emitted per unit area
on the sky as a fn
of angular dist R
and a z. angle ϕ
from G.C.



- Does not depend on ϕ :

$$I(R, \phi) \rightarrow I(R)$$

- Can think of

$$I(R) \equiv \langle I(R, \phi) \rangle \sim \frac{1}{N} \int_0^{2\pi} I(R, \phi) w(\phi) d\phi$$

- Radially averaged intensity of the surface brightness

$$[I(R)] = \frac{M_R}{(\text{arcsec})^2}$$

$$I(R) = -2.5 \log(I(R)) + zpt$$

$\mu \equiv$ magnitudes per square arcsec
 $\equiv$ app. mag. of the eq. total light obs.
 in a square arcsec

$$\mu = -2.5 \log(I(R))$$

$$* I(R) = I_0 e^{-R/R_h}$$

R_h : exp. scale length

$$R_h \approx 1-10 \text{ kpc}$$

$$(R_h)_{MW} \approx 2.5 \text{ kpc}$$

$$L = \int_0^{2\pi} d\phi \int_0^\infty I(R, \phi) R dR$$

face on: $I(R, \phi) \rightarrow I(R)$

$$L = 2\pi \int_0^\infty I_0 e^{-R/R_h} R dR$$

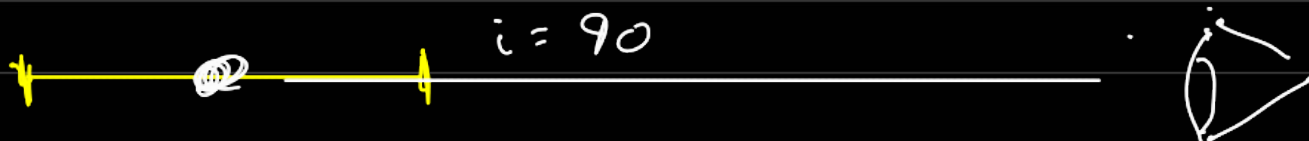
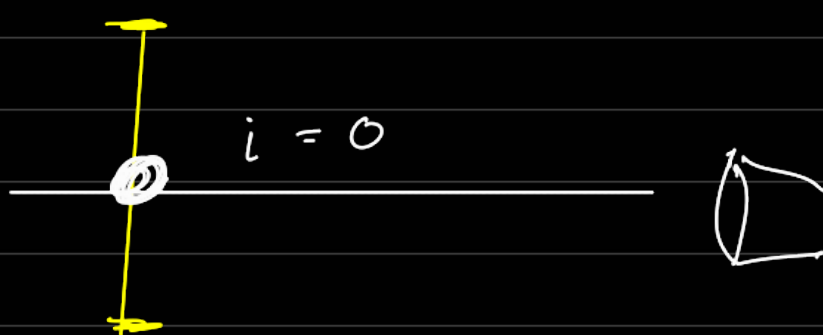
$$L = 2\pi I_0 R_h^2$$



$$L_{\text{obs}} = L \cos(i)$$

$$L_{\text{true}} = \frac{L_{\text{obs}}}{\cos(i)}$$

$$\cos(i) \approx b/a$$



* Late type: *Spiral* *

Early type: *elliptical* *

B-band: 21.65 mag/sq (Freeman's Law)

• Not universal

Low-Surface-Brightness (LSB)

Surface Brightness profiles of Ellipticals:

* de Vaucouleurs $R^{1/4}$ profile:

$$I(R) = I_e \exp \left[-7.67 \left[\left(\frac{R}{R_e} \right)^{1/4} - 1 \right] \right]$$

R_e : radius that encloses half of the total light

I_e : S.B. at R_e

$$L = \int_0^{2\pi} d\phi \int_0^\infty dR R I(R) \quad \Bigg| \quad L = 7.22 \pi I_e R_e^2$$

Sersic Profile:

$$I(R) = I_e e^{-b \left[\left(\frac{R}{R_e} \right)^{1/n} - 1 \right]}$$

b : chosen such that R_e includes half the light.

$n=1$: exp. law

$n=4$: de Vaucouleurs

* Non-parametric Morphology Measurements

$n=2$: Disk like

$n=4$: early type (Elliptical)

CAS-system:

C : Concentration

$$C \equiv 5 \log \left(\frac{r_{80}}{r_{20}} \right)$$

r_{80} : radius that contains 80% of total light

r_{20} : " "

- Tells us how much light is found in bulk.

ellipticals: $4 < C < 5$

Disk: $3 < C < 4$

A : Rotating image 180° subtracting from original

$$A = \frac{|I - R|}{I}$$

I : original image

R : rotated image

$$0 < A < 1$$

$A=0$: totally symmetric

$A=1$: totally A-symmetric

S : Subtracting original from smoothed or blurred version

$$S \equiv \frac{I - B}{I}$$

$$0 < S < 1$$

$S=0$: smooth

$S=1$: clumpy

Contents of Galaxies:

- Ell. star based fast
- Spiral content

- red in E & in spiral bulges $\Rightarrow$ old metal rich
- Spiral arms contain most

Dynamics:

- D.M. in other galaxies:
- Head on spiral

$$\frac{\Delta\lambda}{\lambda} = \frac{V}{c}$$

infer

$$\frac{\Delta\lambda c}{\lambda}$$

$$V(R) \sim$$

- look at different bands to find a best avg. fit.

• HI: 21cm (outer)

• H α or CO in inner: provide a finer resolution

Kepl. orbits:

$$M(<R) = \frac{R V^2}{G}$$

* Take $\left[\frac{M}{L}\right]$ ratios as we did for MW

* Decline in stellar light as $R \rightarrow \infty$ would imply $V \rightarrow$ decline but actually stays constant

* Possible in DM halo from ...

$$\rho(R) \sim \frac{1}{R^2}$$

$$\rho(R) \sim \frac{M}{R^3} = \frac{1}{R^2} \Rightarrow M(R) \sim R$$

- Diverge unless $\exists$ an edge of the D.M. Halo
- Spirals: $\left[\frac{M}{L}\right] \sim 4-20$

Stellar Motions & mass of ellipticals:

- Stars in E do not follow an ordinary pattern/rotation
- Don't have much 'cold gas' $\rightarrow$ highly energetic Vir. flux:

$$2K + U = 0$$

$$K = \frac{1}{2} \sum_i m_i v_i^2 = \frac{1}{2} M_{gal} \langle v^2 \rangle$$

$$\sigma^2 = \sigma_{obs}^2 - \sigma_{intrinsic} \quad \text{(stellar absorption lines)}$$

$$\langle v^2 \rangle = \langle v_x^2 + v_y^2 + v_z^2 \rangle = 3 \langle v_r^2 \rangle = 3 \sigma_r^2$$

$$K = \frac{1}{2} M_{gal} 3 \langle \sigma_r^2 \rangle = \frac{3}{2} M_{gal} \sigma_r^2$$

σ_r : Measured line of sight dispersion

$$K = \frac{3}{2} M_{gal} \sigma_r^2$$

Potential:

$$U = - \frac{GmM}{r}$$

$$\Rightarrow dU = - \frac{GM(r)dm}{r}$$

$$dm = 4\pi r^2 \rho(r) dr$$

$$dU = - G M(r) (4\pi r^2 \rho(r)) dr$$

$$dU = - \int_0^r G M(r) 4\pi r^2 \rho(r) dr$$

$$\rho = \text{const.}$$

$$M = \rho \frac{4}{3} \pi r^3$$

$$= - \rho^2 G \frac{16 \pi^2}{3} \int_0^R r^4 dr$$

$$= - \frac{(4\pi)^2}{3} G \rho^2 \frac{R^5}{5} \quad / \quad M_{\text{gal}} = \frac{4}{3} \pi \rho R^3$$

$$U = - \frac{3}{5} \frac{G M_{\text{gal}}^2}{R}$$

Sphere

$$U = - \alpha \frac{G M_{\text{gal}}^2}{R}$$

(more general)

$$\alpha \approx 1 \quad | \quad 2U + U \Rightarrow 3 M_{\text{gal}} \sigma_r^2 = \frac{G M_{\text{gal}}^2}{R}$$

$$\sigma_r^2 \approx \frac{5 R \sigma_r^2}{5 R \sigma_r^2}$$

$$M_{\text{gal}} \approx \frac{G}{\sigma_r^2} \quad (\text{spherical})$$

$$\sigma_r \approx 200-300 \quad | \quad \left[\frac{M}{L} \right] \approx 7-14$$

- DM exists but is not prominently identified using $\left[\frac{M}{L} \right]$

Galaxy line width luminosity

- Correlate L_{gal} to dynamics:

$$L \propto V_{\text{max}}^4$$

(Tully-Fisher)

- Comes from 21 cm (integrated) lines in spiral gal.
 $\rightarrow$ Measures max velocities within the gal.

$$M_{\text{gal}} \sim 2 V_{\text{max}}^2 \left(\frac{R}{G} \right)$$

$$\Delta W \sim L v_{\max} \sin(i)$$

Ellipticals: $L \propto \sigma_r^4$ (Faber-Jackson)

Simple theory: $V^2 \propto \frac{M}{R}$ | M : mass-to-light

$$\mu(R) = -2.5 \log(I(R)) + C$$

• Use distance modulus:

$$\mu(R) = -2.5 \log(I(R)/I_0) + M_0 + 21.572$$

$$L \propto \frac{R V^2}{\mu}$$

$$L_{\text{tot}} = 7.22 \pi R_e^2 I_e \quad (\text{Elliptical})$$

$$L_{\text{tot}} = 2\pi R_h^2 I_0 \quad (\text{Spiral})$$

I_0 : Central Surface Brightness

I_e : effective " "

R_e, R_h : characteristic scale lengths

* If all have same profile:

$$L \propto I_0 R^2 \Rightarrow R \propto L^{1/2} I_0^{-1/2}$$

$$L \propto \frac{R V^2}{\mu} \propto \frac{L^{1/2} I_0^{-1/2} V^2}{\mu}$$

$$L \propto \frac{V^4}{\mu^2 I_0}$$

$$L \propto V^4$$

(i) I_0 constant

(ii) μ constant

(iii) Roughly all have same profile shape

• Parameter space: $(R_e, \sigma, \langle I_e \rangle)$
(fundamental - Plane)

Ellipticals:

$$\log(R_e) \propto 2 \log(\sigma) - \log(\langle I_e \rangle)$$

• Real: $\log(R_e) \propto 1.24 \log(\sigma) - .82 \log(\langle I_e \rangle)$

• D.H. comes from tilt of plane

Ellipticals: $2K + U = 0$

$$\frac{1}{2} M_{gal} \langle v^2 \rangle = \alpha \frac{G M_{sc}^2}{R}$$

α : depends on model/system

$$\frac{1}{2} \frac{\sigma_r^2 R}{\alpha G} = M_{gal}$$

$$L_{tot} \approx (\text{const}) R^2 \langle I_e \rangle$$

$$\left(\frac{M}{L} \right) \equiv \mu$$

Divide by L : both sides

$$\frac{\sigma_r^2 R}{L} = \mu \Rightarrow \beta \frac{\sigma_r^2}{R^2 \langle I \rangle} = \mu$$

• Assume μ constant:

$$\frac{\sigma_r^2}{R \langle I \rangle} = \omega$$

$$2 \ln(\sigma_r) = \ln(R) \ln(\omega) + \ln(\langle I \rangle)$$

$$\ln(R) = 2 \ln(\sigma_r) - \ln(\langle I \rangle) + C$$

$$\ln(R_e) \propto 2 \ln(\sigma_r) - \ln(\langle I \rangle)$$

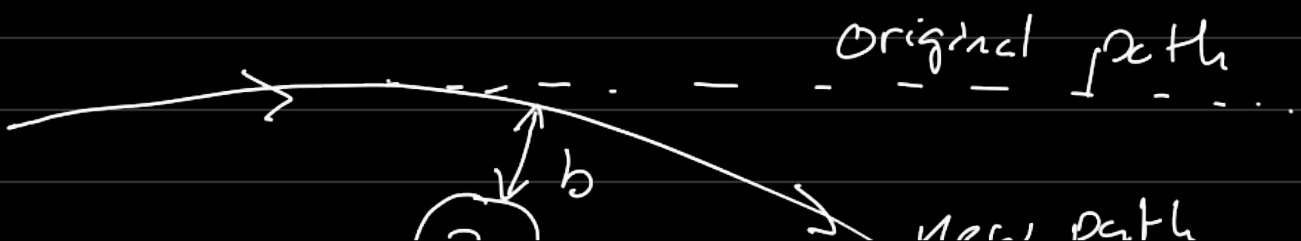
Collisional Relaxation:

• Consider stars in system w mass m .



● : mass: m
vel.: v

→ May interact with another star gravitationally.
This is a gravitational collision.



Q: How close must 1 & 2 be?

A: If the deflection of the trajectory of the star involves a change in momentum of the original star at least as large as its original mom.
 $\Rightarrow$ "Collision"

$$\Delta p_1 \approx p_1 \approx mv$$

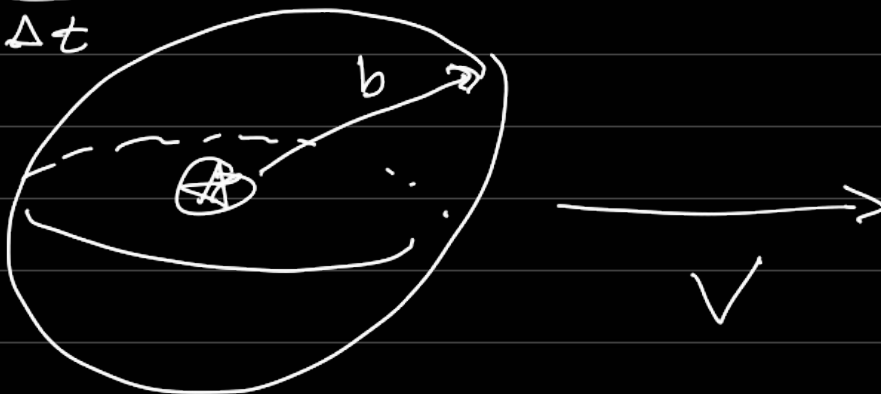
$$F_{12} = \frac{Gm^2}{b^2} = \frac{dp}{dt} \approx \frac{\Delta p}{\Delta t} \approx \frac{mv}{\Delta t}$$

$$\Delta t \approx \frac{b}{v} \text{ (interaction time)}$$

$$\frac{Gm^2}{b^2} \approx \frac{mv^2}{b} \Rightarrow b \approx \frac{Gm}{v^2}$$

* Moving star sweeps out a volume (in unit time) of

$$\frac{\overline{V}}{\Delta t} \approx 4\pi b^2 v$$



$$n \equiv \frac{N}{V}$$

$$\frac{N}{\Delta t} = \frac{nV}{\Delta t} = n \left(\frac{V}{\Delta t} \right) = n b^2 V$$

$$\boxed{\frac{N}{\Delta t} = 4\pi n b^2 V} \quad \text{Collisions / time}$$

* For 1 collision $N = 1$

$$\frac{1}{\tau_{rel}} = n b^2 V 4\pi$$

$$\Rightarrow \boxed{\tau_{rel} = \frac{1}{n b^2 V 4\pi}}$$

replace $b = Gm/V^2$

$$\boxed{\tau_{rel} = \frac{V^3}{4\pi G^2 m^2 n}}$$

(Typical time between collisions)

* For v in $[km/s]$ & $m \approx M_\odot$

$$\boxed{\tau_{rel} \approx 10^{10} \frac{v^3}{n} \text{ yr}}$$

	$v [km/s]$	$n [pc^{-3}]$	$\tau_{rel} [yr]$
Galaxy	100	0.1	10^{17}
open clust.	0.5	1	10^9
glob clust	10	10^3	10^{10}

* age of universe $\sim 10^{10} \text{ yr}$

→ violent collisions: Contacting gas

* Star moving w vel. $|\vec{v}|$ in a stellar system of size R

$$\tau_{cross} = \frac{R}{v}$$



$$\frac{\tau_{rel}}{\tau_{cross}} \approx \frac{4\pi G^2 m^2 n}{(R/V)}$$

$$\approx \frac{V^4}{4\pi G^2 m^2 n R}$$

* If the system is in Virial eq. then
 $2K + U = 0$

$$M_{tot} \langle v \rangle^2 = \frac{G M_{tot}^2}{R}$$

$$\langle v \rangle^2 \approx \frac{G M_{tot}}{R}$$

$$M_{tot} = N m : \quad \langle v \rangle^2 = \frac{G N m}{R}$$

• plugging this in:

$$\frac{\tau_{rel}}{\tau_{cross}} \approx \frac{N^2}{4\pi n R} \approx N$$

$$\tau_{rel} \approx N \tau_{cross}$$

* Analytic sol. has $\sim \frac{1}{\ln(\Delta)}$

